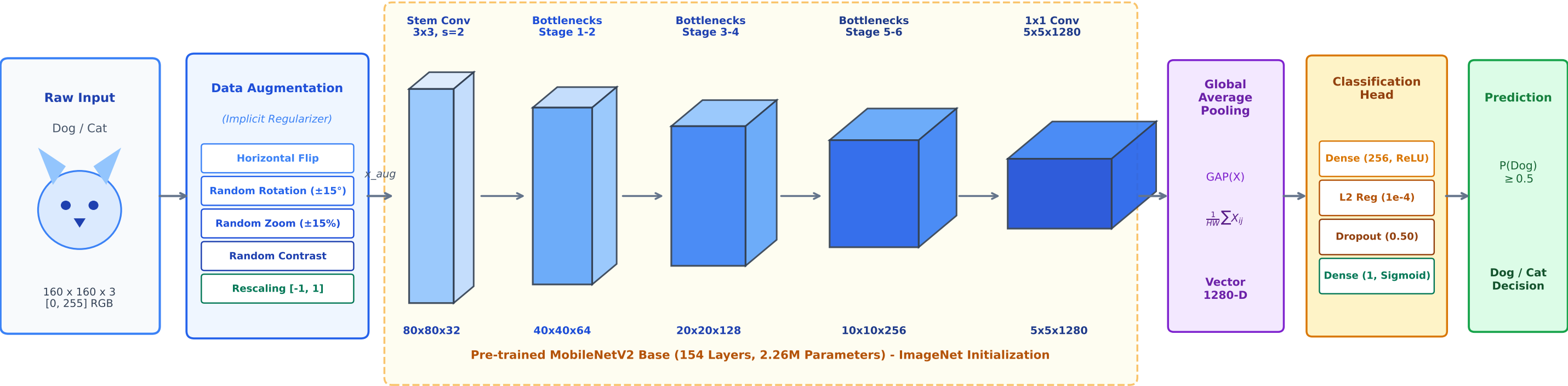


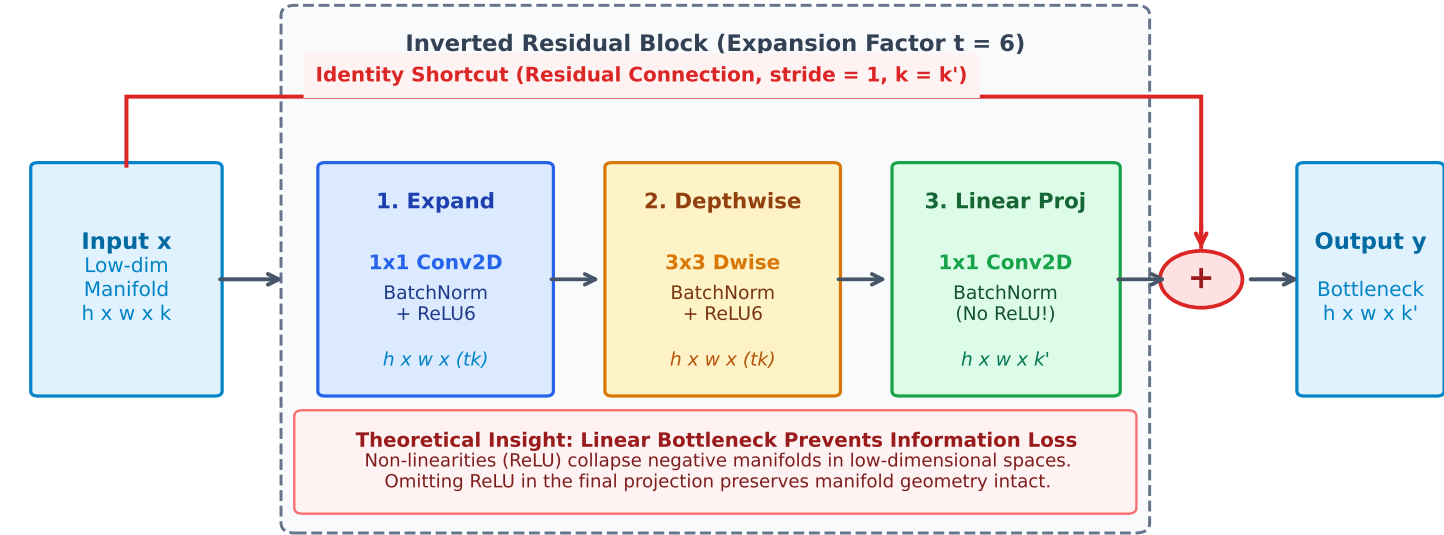
Panel (a): End-to-End Transfer Learning & Tensor Dataflow Pipeline (MobileNetV2 Backbone)

Sequential spatial downsampling, dynamic online augmentation, depthwise feature factorization, and GAP classification head



Panel (b): MobileNetV2 Inverted Residual Block with Linear Bottleneck

Expansion (1x1) → Depthwise Conv (3x3) → Linear Projection (1x1) with Identity Shortcut



Panel (c): Architectural Efficiency & Complexity Synthesis

Custom Scratch CNN vs. MobileNetV2 Transfer Learning Structural Comparison

Architectural Dimension	Custom Scratch CNN	MobileNetV2 (Proposed)
Base Backbone Layers	4 Conv Blocks (Scratch)	154 Inverted Residuals (ImageNet)
Total Model Parameters	4,817,537 (~4.82 M)	2,586,177 (~2.59 M) [-46.3%]
Feature Map Transition	Flatten: 20x20x256 → 102,400	GAP: 5x5x1280 → 1,280 [-98.7%]
Dense Head Parameters	26,214,657 (Extreme Overhead)	328,193 (Compact & Regularized)
Computational Primitives	Standard 2D Convolution	Depthwise Separable (~8.5x FLOPs ↓)
Validation Accuracy	71.49% (High Overfitting Gap)	97.47% (Phase 2 Fine-Tuned)
Commercial Precision	71.85% (Unacceptable for Launch)	97.59% (Exceeds 90-95% Target)
Edge Latency (INT8)	64.2 ms / frame	15.3 ms / frame (65.4 FPS)

Key Architectural Insights: Why MobileNetV2 Dominates Custom Scratch CNN

- Global Average Pooling (GAP) eliminates 98.7% of transition parameters, preventing dense overfitting.
- Depthwise Separable Convolutions deliver ~8.5x theoretical speedup for real-time video streaming.
- Two-phase transfer learning utilizes rich ImageNet semantic priors, boosting precision from 71.85% to 97.59%.